

SIMATS ENGINEERING

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EXPERIMENT 15

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MACHINE LEARNING

Implement Iris Flower Classification using Naive Bayes classifier

Code

```
# Iris Flower Classification using Naive Bayes

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, confusion_matrix, ConfusionMatrixDisplay
import matplotlib.pyplot as plt

# Load Iris dataset
iris = load_iris()

X = iris.data
y = iris.target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Create Naive Bayes classifier
model = GaussianNB()

# Train the model
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Accuracy
accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)
print("Accuracy (%):", accuracy * 100)

# Confusion Matrix
```

```

cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix:")
print(cm)

ConfusionMatrixDisplay(
    confusion_matrix=cm,
    display_labels=iris.target_names
).plot()

plt.title("Iris Flower Classification using Naive Bayes")
plt.show()

```

OUTPUT:

```

*** Accuracy: 1.0
    Accuracy (%): 100.0

Confusion Matrix:
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]

```



